github.com/ttttttris

Jiayi Tian

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Focus on Efficient LLM Training & Inference, Efficient CoT Reasoning via Algorithm-System Co-Design.

EDUCATION

University of California, Santa Barbara, *Ph.D. in Computer Engineering* | CA, USA **3.93/4.0** Fall 2023 - ongoing
University of California, Santa Barbara, *M.S. in Computer Engineering* | CA, USA **3.93/4.0** Fall 2023 - Fall 2025
Nanjing University, *B.Eng. in Electrical Engineering* | China **4.51/5.0** Fall 2019 - Fall 2023

INDUSTRIAL EXPERIENCE

Intel Corporation, *Research Intern 2025* | Hillsboro, OR Jun 2025 – Sep 2025

- Proposed SkipKV, a training-free KV-cache compression framework featuring sentence-level selective eviction and dynamic generation control for efficient chain-of-thought (CoT) reasoning in multi-batch settings.
- Designed a semantic similarity metric to evict redundant sentences from the cache, improving reasoning coherence.
- Introduced a dynamic activation-steering mechanism to enable concise and stable inference.
- Demonstrated strong results on long-reasoning tasks (e.g. AIME24, LiveCodeBench) with LRMs: up to 26.7% higher accuracy, 1.6× shorter generation, and 1.7× higher throughput vs. SOTA under equal compression.
- Resulting paper accepted to MLSys, 2026.

Intel Corporation, *Research Intern 2024* | Hillsboro, OR Jun 2024 - Sep 2024

- Proposed a tensor-compressed Transformer training accelerator on FPGA, optimizing compute ordering, dataflow, and memory allocation for LLMs.
- Designed a bidirectional tensor contraction scheme enabling substantial reduction in intermediate memory and compute cost during long-sequence training and inference.
- Built an HLS-based training engine achieving up to 51× memory efficiency and 4× energy efficiency compared with an Nvidia RTX 3090 GPU.
- Resulting paper accepted to IEEE TCAD, 2025.

AMD-Xilinx Technology, *Co-Op/Intern* | Beijing, China Jun 2023 - Sep 2023

- Developed a C++/HLS Transformer training framework with custom tensorized linear layers and nonlinear operations for LLM acceleration, achieved 30× ∼ 52× saving in model size for end-to-end Transformer training.

SKILLS & RESEARCH INTERESTS

Languages & Tools	Python, PyTorch, Huggingface, vLLM, C/C++, High-level Synthesis (HLS)
ML & NLP	Efficient Large Language Models (LLMs) Training/Inference, Efficient Large Reasoning Models (LRMs) (Model Compression, KV Cache Compression, Pruning, Low-rank decomposition, Early Exit, Knowledge Distillation, Quantization, Speculative Decoding), Multi-Agent System Design

SELECTED PUBLICATIONS & PREPRINTS

RankGuide: Tensor-Rank-Guided Routing and Steering for Efficient Reasoning [paper]
Jiayi Tian, Yupeng Su, Ryan Solgi, Souvik Kundu, Zheng Zhang, arxiv, 2026.

SkipKV: Selective Skipping of KV Generation and Storage for Efficient Inference with Large Reasoning Models [paper] [code]
Jiayi Tian, Seyedarmin Azizi, Yequan Zhao, Erfan Baghaei Potraghloo, Sean McPherson, Sharath Nittur Sridhar, Zhengyang Wang, Zheng Zhang, Massoud Pedram, Souvik Kundu, accepted to MLSys, 2026.

FLAT-LLM: Fine-grained Low-rank Activation Space Transformation for Large Language Model Compression [paper] [code]
Jiayi Tian, Ryan Solgi, Jinming Lu, Yifan Yang, Hai Li, Zheng Zhang, accepted to Findings of EACL, 2026.

Ultra Memory-Efficient On-FPGA Training of Transformers via Tensor-Compressed Optimization [paper]
Jiayi Tian, Jinming Lu, Hai Li, Xiangwei Wang, Cong (Callie) Hao, Ian Young, Zheng Zhang, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2025.

BEBERT: Efficient and robust binary ensemble BERT [paper] [code]
Jiayi, Tian, Chao Fang, Haonan Wang, and Zhongfeng Wang, IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2023.

RESEARCH PROJECTS

Multi-Agent System Acceleration via Speculation Apr 2026 – ongoing

- Developed speculative tool-calling baselines for human-agent interaction benchmarks (e.g., τ-bench).
- Designing a multi-agent system with speculation to improve inference efficiency and task accuracy.

Tensor-Rank-Guided Steering and Routing for High-performance SRM Reasoning Jan 2026. – Mar 2026.

- Proposed RankGuide, a framework that leverages tensor-rank signals from hidden states to accelerate LRMs reasoning, which can yield up to 1.75$\times$ and 1.36$\times$ latency benefit compared to LRM and SoTA collaborative inference framework, while maintaining or improving the accuracy.
- Designed a tensor-rank scoring metric on step-level hidden states to detect low-quality reasoning steps and selectively route them to larger models, improving the accuracy–latency trade-off.
- Developed a rank-based calibration pipeline that filters low-rank samples to construct high-quality steering vectors for inference-time hidden-state intervention, encouraging concise and stable reasoning.

Structural Pruning for Efficient LLM Inference via Low-rank Decomposition Aug 2024 - May 2025

- Developed FLAT-LLM, a training-free, fine-grained compression method that leverages the low-rank structure of the activation space to transform and compress the model weights.
- Introduced a novel training-free rank selection algorithm that allocates ranks using a greedy redistribution strategy and can be integrated with existing low-rank LLM compression pipelines.
- Achieved strong performance on LLaMA-2, 3 and Mistral models with minimal calibration overhead (within minutes), validated across language modeling and downstream tasks.

Binary-Quantized Ensemble LLM for Fast and Robust Language Model Inference Apr 2021 - Jun 2023

- Developed BEBERT, a novel quantization-ensemble strategy enabling efficient and accurate 1-bit BERT inference.
- Leveraged efficient knowledge distillation strategy for high training efficiency.
- Achieved 13$\times$ model size reduction and 15$\times$ compute savings over standard BERT with minimal accuracy loss.
- Proposed early-exit inference variant, further cutting compute by 20% $\sim$ 40% on GLUE benchmark.